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Behavioural response of bivalve molluscs to calcium hydroxide

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ABSTRACT

Blue mussels (*Mytilus edulis*) that are cultivated in the marine area around Prince Edward Island, Eastern Canada, are susceptible to the heavy biofouling of their shells by an invasive solitary tunicate, *Styela clava*, which rapidly proliferates. To mitigate this issue, mussel farmers periodically lift the longlines on which the mussels are suspended out of the water to spray a highly alkaline (~12.7 pH units) calcium hydroxide solution onto fouled individuals. Here, we tested the hypothesis that calcium hydroxide exerts behavioural stress on mussels and other bivalves. Field surveys revealed that the alkalinity of the seawater in the vicinity of longlines increased (9.3–11.7 pH units) immediately after treated mussel sleeves were returned into the water column. Thereafter, pH values declined rapidly, and met federal water quality guidelines (7.0–8.7 pH units) within 3.1 ± 0.5 min (range 0.3–10.5 min, $n = 31$ sleeves). Cultivated mussels challenged to both emersion and calcium hydroxide closed their valves for 14.0 ± 3.3 min ($n = 18$) compared to 6.5 ± 1.6 min ($n = 17$) by control mussels (emersion only). We subsequently assessed how three benthic bivalve species (*M. edulis*, *Crassostrea virginica* (eastern oyster) and *Argopecten irradians* (bay scallop)) respond to weak (pH ≤ 9.2) but sustained (3-h daily for 3 days) alkalinity conditions. All three species consistently responded by completely or partially closing their valves. However, all behavioural responses were short-lived (0.2–4.7 h), and were generally confined to the treatment period. In conclusion, spraying calcium hydroxide onto cultivated mussels has limited impact on seawater alkalinity and the behaviour of nearby bivalves.

1. Introduction

Limestone, or calcium carbonate ($CaCO_3$), is a naturally-occurring sedimentary rock that consists of calcium, magnesium carbonate and/or dolomite, in addition to small amounts of other minerals. When limestone is heated to temperatures >1000 °C it loses carbon dioxide (CO_2), and is converted to calcium oxide (CaO). The process of burning limestone to produce granulated calcium oxide has been practiced for centuries for use in agricultural purposes, such as reducing soil acidity. Among mariculturists, there is also a long history of using calcium oxide to control microorganisms in fish/shrimp/prawn ponds (Boyd and Fast, 1992; Gräslund and Bengtsson, 2001; Shahidul Islam et al., 2004), predatory starfish in oyster beds (Wood, 1908) and sea urchins in kelp beds (Bernstein and Welsford, 1982).

Calcium hydroxide ($Ca(OH)_2$) is known as hydrated lime, flaked lime or builders lime, and is prepared by hydrating calcium oxide in a process that creates a more stable form of lime for storage purposes. Calcium hydroxide has been used for decades to remove the predatory sea star (*Asterias vulgaris*) from oyster (*Crassostrea virginica*) and mussel (*Mytilus edulis*) seed collection systems (Barkhouse et al., 2007; MacKinnon et al., 1993). It is also commonly applied to bivalve stocks prior to their transfer from one embayment to another. This process is aimed at controlling the spread of various marine pests, such as barnacles. However, over the past decade, calcium hydroxide has also been increasingly used to control invasive fouling tunicates, particularly the solitary clubbed tunicate *Styela clava*. There are several reasons for controlling solitary tunicate populations in bivalve farms. For instance, solitary tunicates are efficient filter-feeders that compete with bivalves for limited particulate food resources; consequently, reducing the capacity for coastal systems to produce bivalves (Comeau et al., 2015). In addition, the rapid proliferation of these tunicates results in massive crop loss, because their weight breaks the byssal thread attachment of mussels, causing them to fall to the bottom where they lose their commercial value (Davidson et al., 2016). High-pressure water is usually used to eradicate soft-bodied tunicates, such as *Ciona intestinalis* (Ramsay, 2014). However, *S. clava* has a leather-like tunic that cannot be perforated easily using this method. To date, the most efficient strategy to control the on-site infestation of *S. clava* involves the lethal application of calcium hydroxide (Ramsay et al., 2014).

In the mussel industry, growers usually apply calcium hydroxide in late summer/early autumn after *S. clava* has settled onto the mussel sleeves and has reached a length of 20–25 mm. At this point, calcium hydroxide is dissolved in large onboard tanks to create a saturated

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(~4%) and highly alkaline (~12.7 pH units) lime solution that contains approximately 40 g of calcium hydroxide per 960 ml of seawater (Ramsay et al., 2014). The solution is manually sprayed onto tunicate-infested mussel sleeves as they are lifted from the water using a hydraulic crane (Fig. 1b, c). Following a short period of air exposure (45–90 s) that induces the mortality of *S. clava*, the treated sleeves are slowly re-immersed into the water. In recent years, some growers have developed automated low-pressure spray systems to speed up the treatment process (Fig. 1d). The main components of these automated systems are a mixing tank, a booth with multiple low-pressure nozzles (that hangs over the side of the boat) and a recovery system (to collect unused lime solution). Boats equipped with an automated sprayer system treat a 180 m longline containing 400 sleeves in approximately 30 min, compared to 1 h for manual application practices.

Because calcium hydroxide is a chemical that increases the alkalinity of water, it is important to quantify how it affects the environment and non-targeted organisms. To date, the alkalinity footprint associated to calcium hydroxide application in shellfish farms has not been determined in relation to its duration and spatial boundary. There is also a paucity of published literature on the biological effects of calcium hydroxide, with existing studies primarily focusing on determining lethal exposure levels for the bacterium *Vibrio fischeri*, three-spine stickleback *Gasterosteus aculeatus*, sand shrimps *Crangon septemspinosa* and American lobster *Homarus americanus* (Burridge et al., 2010; Locke et al., 2009). To our knowledge, there have been no studies on the sublethal effects caused by applying calcium hydroxide to mussels in farms. Yet, bivalves warrant attention because they are extremely sensitive to their environment (Jørgensen, 1990). Bivalves filter large amounts of ambient water, resulting in their exposing their unprotected soft-tissues to deleterious substances. Furthermore, mussels respond behaviourally and physiologically to changes in the toxicity of substances in the water (Levinton et al., 2002; Nilin et al., 2012; Tran et al., 2003, 2010).

The present study focused on the use of calcium hydroxide ($Ca(OH)_2$) by mussel farms in the embayments around Prince Edward Island (PEI), Canada, which have been subject to *S. clava* (see Fig. 1a for affected farms). We tested the hypothesis that calcium hydroxide exerts behavioural stress on bivalves, including cultured mussels, when applied to control tunicate fouling in longline mussel farms. Our approach was based on measuring the "pH footprint" in mussel farms, and subsequently challenging bivalves in a controlled environment. The response behaviour was evaluated using high frequency valvometry, which involved closely monitoring the extent of valve opening. Valve activity recording devices have been available since the early 20th century (Hopkins, 1931; Nelson, 1921), and are now being increasingly integrated into evaluations of environmental changes or stressors (Basti et al., 2009; Comeau, 2014; Liao et al., 2009; Tran et al., 2010). The opening of values signals the activation of a complex nervous mechanism involving the heart and adductor muscles (Taylor, 1976). This action results in bivalves exposing themselves to the environment, and exercising metabolically demanding processes; namely, the collection and assimilation of food particles. Thus, atypical valve activity, such as a tendency towards valves closure, was assumed to indicate physiological stress. This study is expected to provide important information on current practices in the mussel industry; specifically, how to mitigate *S. clava* with the minimum environmental damage.

2. Methods

2.1. Field surveys of alkalinity

pH measurements were conducted in the Malpeque Bay area to characterize the calcium hydroxide footprint associated to tunicate-treatment practices. Measurements were conducted opportunistically in 2013–2015, when growers were applying calcium hydroxide to mussels fouled with *S. clava*. During these field surveys, water temperature ranged between 7 and 19 °C and water current velocity ranged between 7.1 and 17.1 cm s^{-1}. The water current was measured by deploying an acoustic Doppler current profiler (ADCP, Workhorse Sentinel 1200 kHz ADCP, Teledyne RD Instruments, Poway, CA, USA) on the seabed; the device pointed towards the surface, and measured the water current at sleeve depth (1.0 to 3.0 m below the surface).

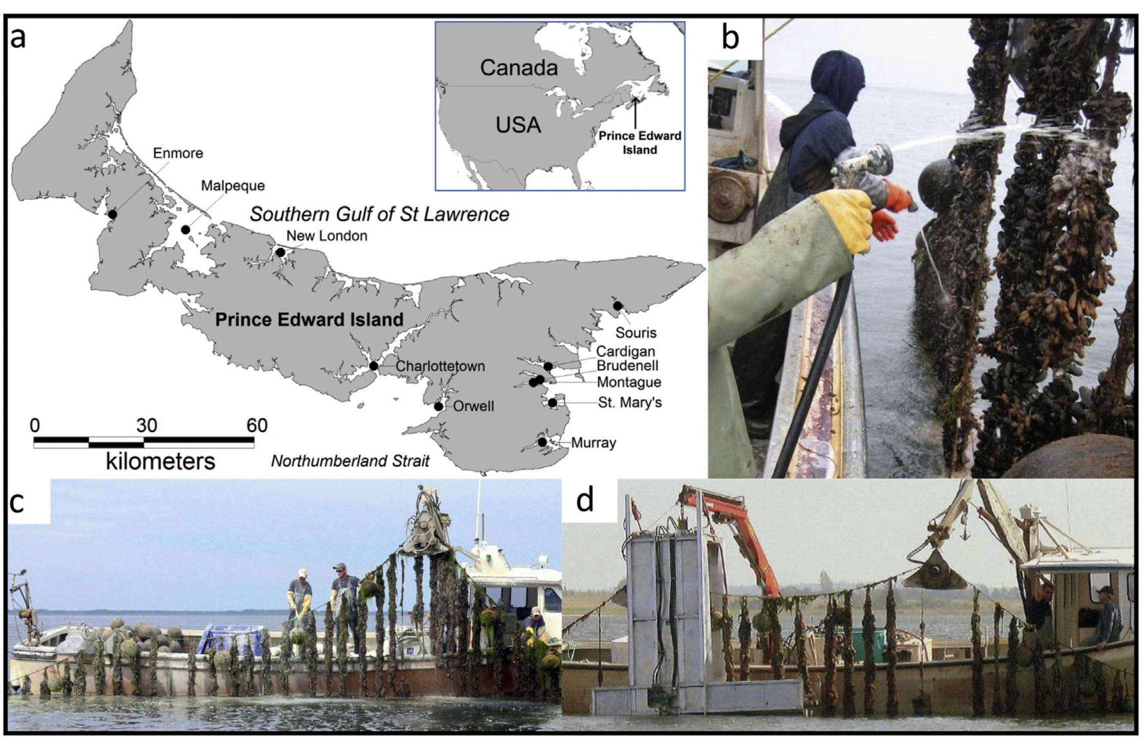

Fig. 1. (a) Distribution of the invasive solitary tunicate, *Styela clava*, in the waters around Prince Edward Island (PEI). (b, c) Mussel sleeves being manually sprayed with a saturated calcium hydroxide solution. (d) Automated/low pressure sprayer system designed to reduce labour costs and calcium hydroxide use.

Seawater pH was measured using two types of automated sondes, the TempHion™ (INW pH/ORP, Instrumentation Northwest, Kent, WA, USA) and YSI multi-parameter 6600 (YSI, Yellow Springs, OH, USA). The sondes were factory-calibrated with an accuracy of ± 0.2 pH units. In the first series of measurements, the sondes were directly attached to fouled mussel sleeves destined for calcium hydroxide treatment. The treatment was applied using four commercial mussel boats; specifically, two replicate boats equipped with manual sprayers and two additional boats equipped with automated devices. Together, these boats treated a total of 31 sleeves for the assessment; 11 using the manual sprayers and 20 using the automated sprayers. Following treatment and re-immersion, the sondes recorded pH every 2 s at a depth of approximately 1.7 m (mid-sleeve).

In a subsequent trial, vertical profiling was conducted at four distances (2, 3, 4, and 5 m) from the treated sleeves, both before and during treatment. For each distance and depth combination, the alkalinity footprint was calculated as the pH that was recorded during treatment minus the pH that was recorded before treatment (ΔpH). The profiling trial was replicated 10 times in the vicinity of a boat equipped with a manual sprayer. From these 10 trials, the mean ΔpH was reported for each distance and depth combination.

Finally, pH sondes were deployed 15 cm above the seabed to characterize the benthic footprint associated to tunicate-treatment practices. Before the calcium hydroxide treatment, divers positioned sondes directly underneath the longlines ($n = 10$) and approximately 8 m away between two adjacent longlines ($n = 13$). Treatments were either performed by a boat equipped with a manual sprayer or an automated sprayer on different occasions.

2.2. Mussel response to pulse alkalinity

In the first experiment, we assessed how cultivated *M. edulis* responded to the levels of calcium hydroxide that they are exposed to during treatment in the field. In September 2014, divers carefully extracted cultivated mussels (shell length 54.5 ± 1.6 mm) from the Malpeque Bay area, and transported them to a field laboratory stationed on Brudenell Pier. Thirty-five mussels were maintained in separate acrylic chambers (1100 ml) that were continuously supplied with natural seawater (temperature ~14.6 °C, salinity ~28 PSU). Following a 1-week acclimation period, the water inflow was halted, and the chambers were allowed to completely drain. This step exposed mussels to air for approximately 25 s, which corresponded to the period of time when fouled sleeves are lifted out of the water using hydraulic cranes and hauler systems. Eighteen mussels were then manually sprayed for 5 s with a saturated (~4%) and highly alkaline (~12.7 pH units) solution containing 40 g calcium hydroxide per 960 ml of seawater. After being sprayed, the mussels were kept emerged for an additional 90 s to simulate the treatment phase that induces the mortality of *S. clava*. Seventeen control mussels received the same treatment; however, the spray only contained seawater. Finally, treated and control mussels were re-immersed by restarting the flow of seawater into the chambers. Water pH was concurrently monitored in each chamber using a series of automated sensors (Vernier PH-BTA, Beaverton, OR, USA) that were connected to a computer. Seawater flow into the chambers was adjusted so that the exponential declines in alkalinity matched the worst-case-scenario previously recorded in the field during the treatment of mussel sleeves by growers in the industry.

Treated and control mussels were connected to a valvometry system that closely monitored valve movement (Comeau, 2014; Nagai et al., 2006). A coated Hall element sensor (HW-300a, Asahi Kasei, Japan) was glued to one valve. A small magnet (4.8 mm diameter × 0.8 mm height) was then glued to the other valve that was located directly below the Hall sensor. The magnet and the Hall element weighed 0.1 and 0.5 g, respectively. For comparative purposes, a 6-mm diameter live barnacle, which is a common epibiont on mussel shells (Doiron, 2008), weighs approximately 0.12 g. The magnetic field (flux density) between the sensor and magnet is a function of the gap between the two valves. The field was recorded in the form of output voltage by dynamic strain recording devices (DC 204R, Tokyo Sokki Kenkyujo Co., Japan). The sampling interval was one measurement $\min^{-1}$. At the end of the monitoring period, the voltage was converted to absolute valve opening (mm), by applying conversion algorithms specific to each sensor assembly. The adductor muscle was severed, and small calibration wedges (1–6 mm in height) were manoeuvred between the two valves at the point farthest from the hinge. The relationships between voltage and wedge height (i.e., valve opening in mm) were confirmed to be strong ($r^2 > 0.92$).

2.3. Bivalve response to sustained alkalinity

In the second experiment, we assessed how three benthic bivalve species respond to the alkalinity levels they may be exposed to during the treatment of tunicates. The experiment focused on two native species, *M. edulis* and *C. virginica*. *Argopecten irradians* (bay scallop) was also included in the experiment because of aquaculture-related interests. In addition, this species is known to be sensitive to environmental variability (Eckman et al., 1989; Mallet and Carver, 2009). Thirty individuals (12 *M. edulis*, 8 *C. virginica* and 10 *A. irradians*) were connected to the valvometry system, and were distributed in four acrylic tanks (1100 ml) supplied with natural seawater (temperature ~17.7 °C, ~ 28 PSU). Each tank contained two to three individuals of each species. After a one-week acclimation period, 30 ml of a saturated (~4%) and highly alkaline (~12.7 pH units) calcium hydroxide solution was added to two of the four experimental tanks. The two remaining tanks (control) received 30 ml additions consisting of seawater only. These additions were repeated every 30 min over a 3-h period. This treatment scheme created six consecutive alkalinity peaks (~9.2 pH units). Experimental exposure was based on benthic pH measurements that were previously conducted in the field, representing the worst case scenario. The scenario was repeated daily over 3 consecutive days.

2.4. Statistics

2.4.1. Field surveys of alkalinity

For the field observations, a mixed model analysis of variance was used to test how sprayer type (Sprayer) affected the pH footprint at the mussel sleeve level. The duration of the pH footprint (termed Time-pH) was assessed as the time required for pH to return to a the federal quality guideline of Canada; namely, 7.0–8.7 pH units (CCME, 1999). Locke (2008) also concluded that most pH-related biological responses occur at pH levels outside the CCME recommended range. The replicated treatment (termed Rep), representing the commercial mussel boat, was set as a random factor. Data were log-transformed because variance was heterogeneous.

$$\mathrm{Time-pH}_{ij} = \mu + \mathrm{Sprayer}_i + \mathrm{Rep}_j(\mathrm{Sprayer}_i) + \varepsilon_{ij}$$

where Time-pH represented the time required for pH to return to the federal quality guideline (<8.7 pH units) following the re-immersion of treated sleeves; μ represented the overall mean of the population; Sprayer represented the type of calcium hydroxide sprayer, where $i = 1$ (manual), 2 (automated); Rep represented the replicated treatment boat, where $j = 1, 2$; and ε represented the error.

We did not attempt to evaluate how sprayer type affected the benthic footprint, because there were no replicates relating to the application method (manual vs automated) in this trial. The analysis was restricted to a comparison between two benthic locations relative to treatment: directly underneath a longline ($n = 10$) and approximately 8 m between two longlines ($n = 13$). Because data transformations failed to stabilise variance, a non-parametric test (Mann-Whitney *U* test) was applied to compare these two benthic locations.

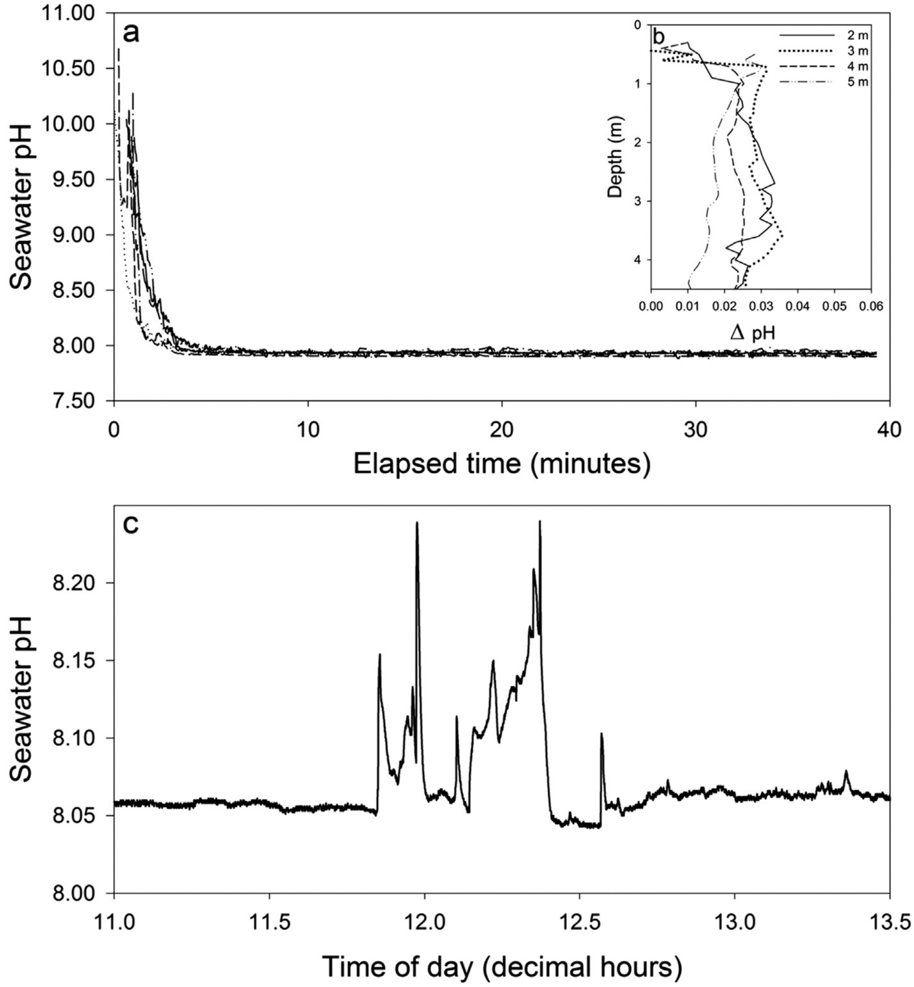

Fig. 2. Field surveys of alkalinity. (a) Exponential decay of alkalinity at the edge of six treated mussel sleeves. pH was measured by attaching sondes to the sleeves. (b) Mean ($n = 10$ trials) ΔpH as a function of water depth and distance from treated sleeves. ΔpH was calculated from two vertical profiles made before and after treatment (at different lateral distances) from the sleeve. (c) Example of pH fluctuation above the seabed during the treatment of nearby mussel sleeves. In this example, pH was measured by a sonde positioned 15 cm above the seabed and directly beneath a longline.

2.4.2. Mussel response to pulse alkalinity

Here, we aimed to assess how cultivated mussels responded to a saturated concentration of calcium hydroxide, mimicking field treatments. The time required for treated ($n = 18$) and control ($n = 17$) mussels to close their valves after emersion (pre-treatment) was analysed, along with the time required for them to re-open their valves following re-immersion (post-treatment). The Mann-Whitney *U* test was used, because transformations failed to stabilise variances.

2.4.3. Bivalve response to sustained alkalinity

A relative valve-opening metric was computed as the per cent of the maximal recorded opening amplitude specific to each individual. The metric was then partitioned into 10 ranges from zero to 100% amplitude. Per cent occurrence was calculated as the number of observations in a specified range (e.g. 0–10% amplitude) divided by the total number of observations (Tran et al., 2010). Statistical analyses were restricted to the lowest range of valve opening amplitude (0–10% amplitude), because we aimed to determine whether calcium hydroxide changed the prevalence of complete valve closure or near closure, which indicate avoidance behaviour. A mixed model analysis of variance (SPSS version 20, procedure GLM) was used to test the effect of calcium hydroxide on per cent occurrence within the 0–10% amplitude valve opening category. The model was summarized as:

$$\mathrm{Occ}_{ijkl} = \mu + \mathrm{TR}_i + \mathrm{DAY}_j(\mathrm{TR}_i) + \mathrm{TK}_k(\mathrm{TR}_i \times \mathrm{DAY}_j) + \varepsilon_{ijk}$$

where Occ represented percent occurrence at the specified range of valve opening (0 to 10% amplitude); μ represented the overall mean of the population; TR represented the experimental treatment (i = 1 [calcium hydroxide], 2 [control]); DAY represented the replicated experiment over

Table 1
Field surveys of alkalinity. Mixed model ANOVA testing for the effect of calcium hydroxide sprayer type (Sprayer) and boat replicates (Rep) on the pH footprint at the mussel sleeve level. The footprint (Time-pH) was defined as the time required for pH to return to federal water quality guidelines (7.0–8.7 pH units).

		Time-pH	
Source	d.f.	MS	Pr > F
Sprayer	2	0.72	0.65
Rep(Sprayer)	2	1.47	<**0.01**
Error	27	0.06	

Bold value highlights statistical significance.

a three-day period ($j = 1, 2, 3$); TK represented the replicated tank ($k = 1, 2$); and ε represented model error. TR was set as the main effect, whereas DAY and TK were set as random effects. Data were rank-transformed because variance was heterogeneous (Levene's test).

A one-way repeated measures MANOVA was performed to determine whether the response to calcium hydroxide was species-specific. The response metric was the amount of time bivalves had their valves completely closed when subjected to sustained alkalinity. In the analysis, replicated measurements over the three-day period were set as the within-subjects factors, whereas species was set as the between-subject factor. Post hoc Bonferroni's multiple comparisons were performed to identify which species responded differently.

All analyses were performed in SPSS v. 20 (IBM SPSS Inc., Chicago). Statistical significance was set at $P < 0.05$. Reported values represent means $\pm$ 1 SE of the mean.

3. Results

3.1. Field surveys of alkalinity

The highest mussel sleeve pH values (9.3–11.7) were recorded immediately after the sleeves were returned to the water column (Fig. 2a). Thereafter, pH rapidly declined and met the federal water quality guideline (7.0–8.7 pH units) within 3.1 $\pm$ 0.5 min (range 0.3–10.5 min, $n = 31$ sleeves). The type of sprayer (manual vs automated) had no significant effect on the time required to meet the water quality guideline (Table 1). However, there was a significant replicate effect, indicating different footprint durations across experimental units (boats).

Alkalinity footprints extended beyond the source points (sleeves). Vertical profiling through the water-column revealed footprints at distances ranging between 2 and 5 m from the treated sleeves (Fig. 2b). At these distances, changes in pH (ΔpH) were consistently positive, indicating a shift towards alkalinity. Alkalinity became apparent at approximately 1 m below the surface, which was consistent with the position of treated sleeves, and progressively declined with increasing distance from the sleeves. The profiles shown in Fig. 2b represent ΔpH means from 10 trials. ΔpH means were consistently below 0.06 pH units; the maximum ΔpH was 0.25 pH units recorded at a distance of 3 m from the treated sleeves.

When growers were treating mussel sleeves, all pH sensors deployed 15 cm above the seabed, including those 8 m away from the treatment point, recorded intermittent alkalinity peaks (Fig. 2c). The amplitude of the peaks was consistently weak. pH increased above ambient levels by 0.02 to 0.48 pH units, with the maximum recorded absolute value being 8.4 pH units. Each of these episodic event lasted between 2.4 and 126.0 min, with a mean duration of 36.8 $\pm$ 8.0 min. The positioning of sensors in relation to treated longlines (underneath vs ~8 m away) had no significant effect on the near-bottom pH signatures, either in terms of signal amplitude (Mann-Whitney *U* test, $P = 0.71$) or duration (Mann-Whitney *U* test, $P = 0.93$).

3.2. Mussel response to pulse alkalinity

On emersion, the laboratory mussels destined for treatment completely closed their valves in 26 $\pm$ 5 s (range 2–147 s, $n = 35$). The time required for valve closure was similar between control and to-be-treated mussels (Mann-Whitney *U* test, $P = 0.92$). Surprisingly, the majority (64%) of mussels had their valves marginally (0.54 $\pm$ 0.03 mm) open when calcium hydroxide was applied and during the subsequent drying period. Following re-immersion, calcium hydroxide-treated mussels re-opened their valves after 14.0 $\pm$ 3.3 min ($n = 18$) compared to 6.5 $\pm$ 1.6 min ($n = 17$) by control mussels (Mann-Whitney *U* test, $P = 0.02$). Post treatment monitoring of the mussels over a 14-d period indicated no mortality.

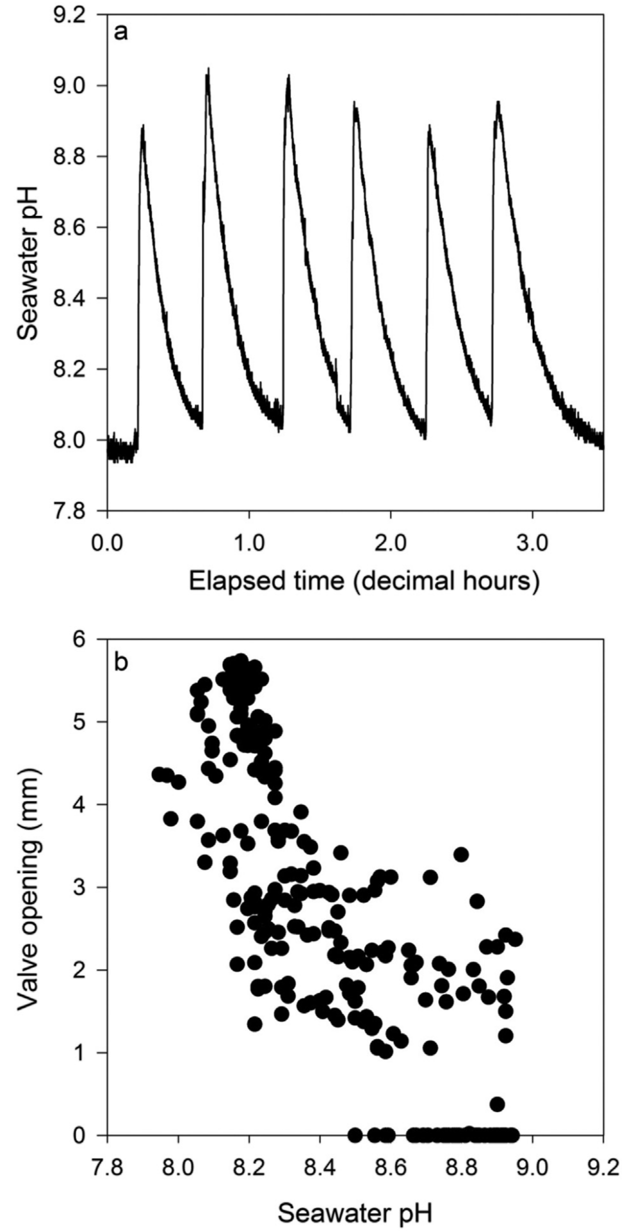

Fig. 3. Results from the sustained alkalinity experiment. (a) Example of calcium hydroxide addition, creating six consecutive alkalinity peaks over a 3-h period. (b) Typical valve opening as a function of increasing alkalinity; data from one *M. edulis* responding to an alkalinity peak event.

3.3. Bivalve response to sustained alkalinity

The typical response of bivalves to the repeated alkalinity increases (Fig. 3a) generally resulted in partial and complete valve closure (Fig. 3b). This observation was further supported by a detailed analysis of the amplitude of valve-opening. Fig. 4 shows the percentage occurrence as a function of valve-opening amplitude (10 ranges from 0 to 100% of maximum amplitude) for calcium hydroxide-exposed (black bars) and control (grey bars) bivalves. For bivalves exposed to calcium hydroxide, the mean occurrence followed a positively skewed normal distribution. The modes of opening amplitudes were most frequently observed in the range of 0–10%, indicating a tendency towards complete valve closure or near closure. This behavioural response was significant for all three investigated species (Table 2), even though the responses varied across experiment replicates for *C. virginica*. The response for all

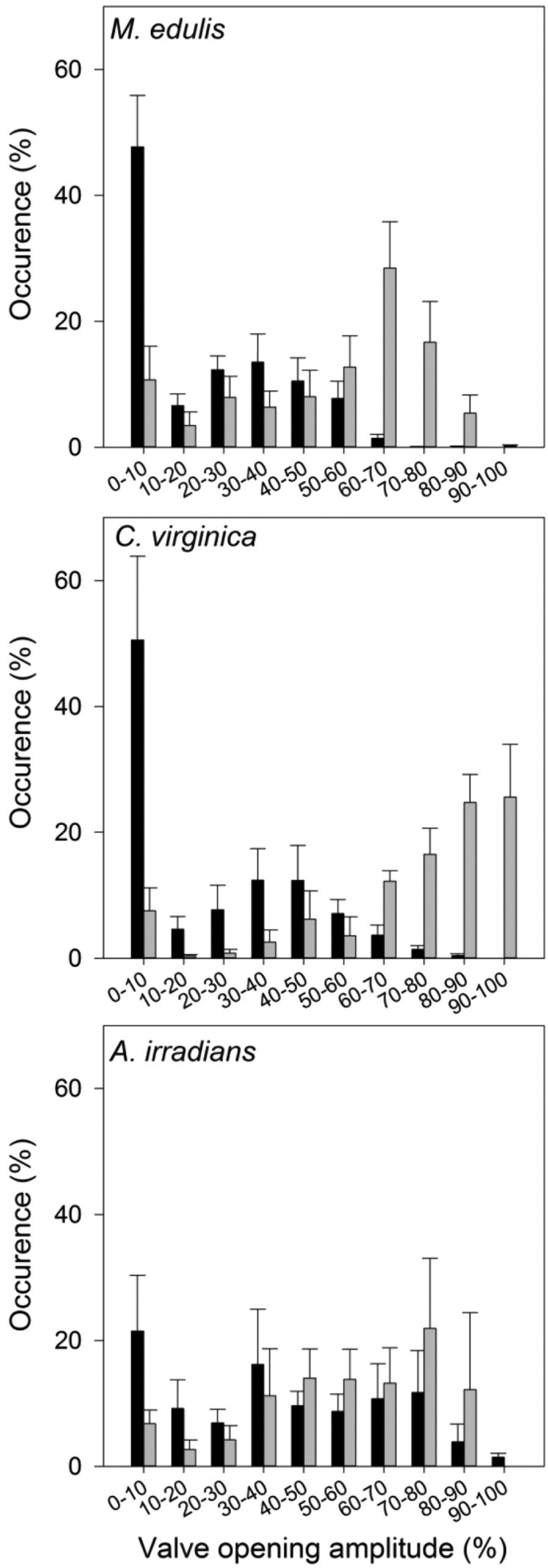

Fig. 4. Bivalve response to sustained alkalinity. Per cent occurrence as a function of valve-opening amplitude (10 ranges from 0 to 100% maximum amplitude) for calcium hydroxide-exposed (black bars) and control (grey bars) bivalves. Valve opening was recorded every minute over a 3-hour treatment period, which was repeated over three consecutive days. Error bars show mean ± standard error.

Table 2
Bivalve response to sustained alkalinity. Mixed model ANOVAs testing for the effect of dissolved calcium hydroxide (TR), repeated treatment (Day), and tank replicates (TK) on the occurrence of valve closure or near closure (<10% opening amplitude).

Source	Species	d.f.	MS	F	*P*
TR	*M. edulis*	2	21,522.44	12.22	**0.02**
	C. virginica	2	18,000.76	18.46	**0.01**
	A. irradians	2	71,282.60	193.88	<**0.001**
DAY(TR)	*M. edulis*	4	1761.65	3.75	0.07
	C. virginica	4	975.22	5.96	**0.03**
	A. irradians	4	367.66	0.60	0.68
TK(TR × REP)	*M. edulis*	6	469.25	0.96	0.48
	C. virginica	6	163.80	0.16	0.98
	A. irradians	6	617.36	0.60	0.75
Error	*M. edulis*	24	490.81		
	C. virginica	12	1050.03		
	A. irradians	18	1085.29		

Bold values highlight statistical significance.
A. irradians was ranked-transformed.

three investigated species was transient, with complete valve closures lasting between 0.2 and 4.7 h.

Fig. 5 shows the amount of time that bivalves kept their valves completely closed when subjected to sustained alkalinity. Repeated measures MANOVA indicated statistically significant differences among species [$F(2,14) = 4.54$, $P = 0.03$]. A post hoc pairwise comparison analysis revealed that on average *M. edulis* remained closed for 1.4 h longer compared to *A. irradians* ($P = 0.04$). Post-treatment monitoring of the mussels over a 14-d period indicated no mortality.

4. Discussion

4.1. Field surveys of alkalinity

The sensors indicated that water pH at the treatment point (sleeves) increased above a recommended environmental guideline (8.7 pH units, CCME, 1999) for 3.1 ± 0.5 min following the re-immersion of treated sleeves. The sensors also detected alkalinity footprints at distances up to 5 m (pelagic) and 8 m (benthic) away from the treated sleeves; however, these footprints were consistently within the CCME limit (8.7 pH units). Surprisingly, the method of application (manual vs automated sprayer) had no significant effect on the duration of the alkalinity footprint at the source point (sleeve). This outcome was unexpected because automated sprayer systems partially recycle calcium

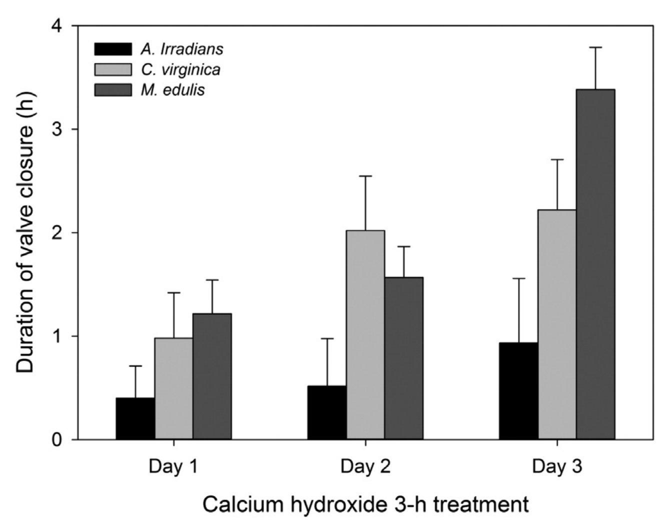

Fig. 5. Bivalve response to sustained alkalinity. Duration of complete valve closure induced by the repeated addition of calcium hydroxide (every 30 min over a 3-h period each day). Error bars show mean ± standard error.

hydroxide, and might release controlled amounts of the alkaline substance into the environment. Only the experimental units (boats) significantly altered the duration of the footprint. Thus, the amount of calcium hydroxide released into the environment appeared to be largely governed by the grower and, perhaps, the level of tunicate infestation at the time of treatment. In addition, multiple confounding environmental variables might have also affected the results, because the different methodologies were tested at different sites and, also, on different days.

The alkalinity footprints described in the present study reflect previous measurements made with hand-held pH meters. Locke et al. (2009) reported that calcium hydroxide application in longline mussel farming increased pH above the 8.7 CCME limit, but only in the immediate vicinity (<1 m) of the treatment sites and for "short" durations. Similarly, Ramsay et al. (2014) concluded that the pH footprint extends up to 2 m from the longline being treated. All other published studies used granular calcium oxide, rather than calcium hydroxide. In the 20th century, calcium oxide pellets were spread on the seabed to eradicate the predatory starfish *Asterias forbesi* from oyster beds, mainly in Long Island Sound (USA). These pellets were spread at high densities (1600–6750 kg ha^{-1}), and dissolved over five to 18 days (MacKenzie, 1977, 1983). Contact of the substance with the sensitive epidermis of the starfish proved lethal (Galtsoff and Loosanoff, 1939; Needler, 1940). Loosanoff and Engle (1942) found that this practise increased water pH by 0.2–0.4 pH units over a few days. In comparison, low concentrations of calcium hydroxide are applied in the PEI longline mussel farms (~70–140 kg ha^{-1}), producing a brief, but accentuated, footprint, increasing pH by several units for just a few minutes.

4.2. Mussel response to pulse alkalinity

Spraying a saturated (~4%) and highly alkaline (~12.7 pH units) calcium hydroxide solution directly onto cultivated *M. edulis* had no lethal effects. However, a behavioural response was detected via valvometry. Following re-immersion, calcium hydroxide-treated mussels completely re-opened their valves after 14.0 ± 3.3 min ($n = 18$) compared to 6.5 ± 1.6 min ($n = 17$) for control mussels. The delay in valve opening was short and was assumed to have no meaningful impact on physiology, because *M. edulis*, like most intertidal bivalves, is able to operate anaerobically for extended periods of time (Bayne, 1973; de Zwaan and Wijsman, 1976). Not treating the mussels would result in much worse consequences, including death from the rapid proliferation of *S. clava*. This tunicate directly attaches to the shells and obstructs the mussel from using the seston and accessing oxygen. For example, if the sleeves are left untreated, the density of *S. clava* reaches up to 3136 individuals per 2-m sleeve (A. Ramsay, Prince Edward Island Department of Agriculture and Fisheries, pers. Comm.). Post-treatment counts of live *S. clava* fall to 627 (Ramsay, 2014) individuals per 2-m sleeve, with this number being comparable to the expected harvestable mussel count, which is 633 ± 64 mussels per 2-m sleeve (Comeau et al., 2015).

4.3. Bivalve response to sustained alkalinity

Ramsay (2014) recorded several fish species in field observations that did not appear to respond when calcium hydroxide was applied above them. However, in the present study, *M. edulis*, *C. virginica*, and *A. irradians* challenged to sustained near-bottom like alkalinity fluctuations (≤9.2 pH units) consistently responded by completely or partially closing their valves. The experimental bivalves were subjected to slightly higher alkaline exposures (maximum of 9.2 pH units for 3 h) compared to those recorded in the benthic environment when mussel growers treat infested sleeves (maximum of 8.4 pH units for 2 h). In their pioneering work on calcium oxide, Loosanoff and Engle (1942) exposed three species (*C. virginica*, *M. edulis*, and *M. mercenaria*) to alkaline environments pH (8.6–9.5) for prolonged periods of time, ranging from 2 to 8 days. The authors reported that shell movements became normal "soon" after the lengthy exposure. We found that the behavioural response was generally confined to the treatment period, and was of short duration, ranging from 0.2 to 4.7 h. Interestingly, this avoidance behaviour was more pronounced for *M. edulis* compared to *A. irradians*. This result indicates that the cultivated species, *M. edulis*, exhibited enhanced sensitivity to calcium hydroxide treatment. The underlying reason for this observation is unknown. It is possible that *M. edulis* closed its valves at a slower rate compared to *A. irradians*, which would have allowed a greater amount of calcium hydroxide to enter its pallial cavity; thus, enhancing avoidance behaviour.

No bivalve mortality was detected at 14 days following the calcium hydroxide exposure in the laboratory. Similarly, exposures to concentrated calcium oxide did not kill nor noticeably injure *C. virginica*, *M. mercenaria*, *M. edulis* and other molluscs (Loosanoff, 1961; Loosanoff and Engle, 1942; Needler, 1940; Shumway et al., 1988). Turner (1970) stated that calcium oxide pellets had no impact on the survival rates of the soft-shelled clam, *Mya arenaria*, and only interrupted normal feeding activity for a "few hours".

5. Conclusions

Spraying calcium hydroxide onto cultivated mussels only minimally impacts seawater alkalinity for a few minutes. Following the re-immersion of treated sleeves, there is a strong (9.3–11.7 pH units) but short-lived (3.1 ± 0.5 min) alkalinity footprint at the edge of the sleeves; the footprint quickly declines, meeting federal water quality guidelines (7.0–8.7 pH units) within a 2-m distance from the sleeves. Bivalves responded by temporarily (0.2–4.7 h) closing their valves. Therefore, calcium hydroxide applied to eradicate tunicates exerts acute, but short-lived, behavioural stress on bivalves, including cultivated mussels. This result supports literature published during the 20th century focused on assessing the effect of granular calcium oxide pellets spread on oyster beds. In the future, it is possible that such buffering substances will be increasingly viewed for their positive ability of combatting ocean acidification, which is detrimental to calcifying marine organisms, even at its current stage (Parker and Ross, 2010; Talmage and Gobler, 2009).

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References

Barkhouse, C.L., Niles, M., Davidson, L.-A., 2007. A literature review of sea star control methods for bottom and off bottom shellfish cultures. Can. Tech. Rep. Fish. Aquat. Sci. 279 vii + 38 p.

Basti, L., Nagai, K., Shimasaki, Y., Oshima, Y., Honjo, T., Segawa, S., 2009. Effects of the toxic dinoflagellate *Heterocapsa circularisquama* on the valve movement behaviour of the Manila clam *Ruditapes philippinarum*. Aquaculture 291, 41–47.

Bayne, B., 1973. Aspects of the metabolism of *Mytilus edulis* during starvation. Neth. J. Sea Res. 7, 399–410.

Bernstein, B.B., Welsford, R.W., 1982. An assessment of feasibility of using high-calcium quicklime as an experimental tool for research into kelp bed/sea urchin ecosystems in Nova Scotia. Can. Tech. Rep. Fish. Aquat. Sci. 968, 1–51.

Boyd, C.E., Fast, A.W., 1992. Pond monitoring and management. In: Lester, A.W.F.J. (Ed.), Marine Shrimp Culture. Elsevier, Amsterdam, pp. 497–513.

Burridge, L.E., Doe, K.G., Ernst, W., 2010. Pathway of effects of chemical inputs from the aquaculture activities in Canada. DFO Can. Sci. Advis. Sec. Res. Doc. 2010/017 vi + 63 p.

CCME, 1999. Canadian Council of Ministers of the Environment. Canadian water quality guidelines for the protection of aquatic life: pH (marine). Canadian environmental quality guidelines, 1999. Canadian Council of Ministers of the Environment, Winnipeg, MB, Canada.

Comeau, L.A., 2014. Spring awakening temperature and survival of sediment-covered eastern oysters *Crassostrea virginica*. Aquaculture 430, 188–194.

Comeau, L.A., Filgueira, R., Guyondet, T., Sonier, R., 2015. The impact of invasive tunicates on the demand for phytoplankton in longline mussel farms. Aquaculture 441, 95–105.

Davidson, J., Landry, T., Johnson, G., Ramsay, A., Queen, C., Quijón, P., 2016. A field trial to determine the optimal treatment regime for *Ciona intestinalis* on mussel socks. Management of Biological Invasions 7, 167–179.

de Zwaan, A., Wijsman, T.C.M., 1976. Anaerobic metabolism in bivalvia (Mollusca) characteristics of anaerobic metabolism. Comparative Biochemistry and Physiology Part B: Comparative Biochemistry 54, 313–323.

Doiron, S., 2008. Manuel de référence de l'ostréiculteur. Ministère d'agriculture, aquaculture et pêches, Fredericton, Nouveau-Brunswick 75 pp.

Eckman, J., Peterson, C., Cahalan, J., 1989. Effects of flow speed, turbulence, and orientation on growth of juvenile bay scallops *Argopecten irradians concentricus* (Say). J. Exp. Mar. Biol. Ecol. 132, 123–140.

Galtsoff, P.S., Loosanoff, V.L., 1939. Natural history and method of controlling the starfish (*Asterias forbesi*, Desor). Bulletin of the Bureau of Fisheries 41, 76–132.

Gräslund, S., Bengtsson, B.-E., 2001. Chemicals and biological products used in south-east Asian shrimp farming, and their potential impact on the environment — a review. Sci. Total Environ. 280, 93–131.

Hopkins, A., 1931. Temperature and shell movements of the oyster (*Ostrea gigas*). Bull. U. S. Bur. Fish. 47, 1–14.

Jørgensen, C.B., 1990. Bivalve Filter Feeding: Hydrodynamics, Bioenergetics. Physiology and Ecology, Olsen & Olsen.

Levinton, J., Ward, J., Shumway, S., 2002. Feeding responses of the bivalves *Crassostrea gigas* and *Mytilus trossulus* to chemical composition of fresh and aged kelp detritus. Mar. Biol. 141, 367–376.

Liao, C.-M., Jau, S.-F., Lin, C.-M., Jou, L.-J., Liu, C.-W., Liao, V.H.-C., Chang, F.-J., 2009. Valve movement response of the freshwater clam *Corbicula fluminea* following exposure to waterborne arsenic. Ecotoxicology 18, 567–576.

Locke, A., 2008. Tabulated Observations of the pH Tolerance of Marine and Estuarine Biota. Canadian Manuscript Report of Fisheries and Aquatic Sciences No. 2857 28 + iv pp.

Locke, A., Doe, K.G., Fairchild, W.L., Jackman, P.M., Reese, E.J., 2009. Preliminary evaluation of effects of invasive tunicate management with acetic acid and calcium hydroxide on non-target marine organisms in Prince Edward Island, Canada. Aquat. Invasions 4, 221–236.

Loosanoff, V.L., 1961. Biology and Methods of Controlling the Starfish, *Asterias forbesi* (DESOR). United States Department of the Interior, Bureau of Commercial Fisheries, U.S. Fish and Wildlife Service, Milford, Connecticut Fishery Leaflet 520.

Loosanoff, V.L., Engle, J.B., 1942. Use of lime in controlling starfish. U S Fish and Wildlife Service Research Report 2, 1–29.

MacKenzie, C.L., 1977. Use of quicklime to increase oyster seed production. Aquaculture 10, 45–51.

MacKenzie, C.L., 1983. To increase oyster production in the Northeastern United States. Mar. Fish. Rev. 45, 1–22.

MacKinnon, C.M., Gallant, R., Gillis, B., 1993. Some Observations on Starfish Predation on Mussel Spat Collectors. P. E. I. Department of Agriculture Fisheries and Forestry Technical Reports Series 208: 28p.

Mallet, A.L., Carver, C.E., 2009. The effect of photoperiod on the reproductive development of the northern bay scallop, *Argopecten irradians irradians*. J. Exp. Mar. Biol. Ecol. 374, 128–133.

Nagai, K., Honjo, T., Go, J., Yamashita, H., Seok Jin, O., 2006. Detecting the shellfish killer *Heterocapsa circularisquama* (Dinophyceae) by measuring bivalve valve activity with a hall element sensor. Aquaculture 255, 395–401.

Needler, A.W., 1940. Use of Quicklime for Killing Starfish on Oyster Grounds. Fisheries Research Board of Canada, Ottawa Oyster Farming Circular 40-FRB-OFCII.

Nelson, T.C., 1921. Report of the Department of Biology of the New Jersey Agricultural College Experiment Station, New Brunswick, N.J. For the Year Ending June 30th 1920. Published by the State, Trenton N.J, pp. 317–349.

Nilin, J., Pestana, J.L.T., Ferreira, N.G., Loureiro, S., Costa-Lotufo, L.V., Soares, A.M.V.M., 2012. Physiological responses of the European cockle *Cerastoderma edule* (Bivalvia: Cardidae) as indicators of coastal lagoon pollution. Sci. Total Environ. 435–436, 44–52.

Parker, L.M., Ross, P.M., 2010. Comparing the effect of elevated pCO2 and temperature on the fertilization and early development of two species of oysters. Marine Biol. (New York) 157, 2435–2452.

Ramsay, A., 2014. High-pressure Water Application to Control the Vase Tunicate and Increase Mussel Productivity. Aqua Info Aquaculture Notes, PEI Department of Agriculture, Fisheries and Aquaculture AIN 23.2014.

Ramsay, A., Gill, K., Morrison, A., MacNair, N., 2014. Hydrated Lime Application by the PEI Aquaculture Industry. PEI Department of Fisheries, Aquaculture and Rural Development Technical Report 253, 26pp.

Shahidul Islam, M., Jahangir Sarker, M., Yamamoto, T., Abdul Wahab, M., Tanaka, M., 2004. Water and sediment quality, partial mass budget and effluent N loading in coastal brackish water shrimp farms in Bangladesh. Mar. Pollut. Bull. 48, 471–485.

Shumway, S.E., Card, D., Getchell, R., Newell, C.R., 1988. Effects of calcium oxide (quicklime) on non-target organisms in mussel beds. Bull. Environ. Contam. Toxicol. 40, 503–509.

Talmage, S., Gobler, C., 2009. The effects of elevated carbon dioxide concentrations on the metamorphosis, size, and survival of larval hard clams (*Mercenaria mercenaria*), bay scallops (*Argopecten irradians*), and Eastern oysters (*Crassostrea virginica*). Limnol. Oceanogr. 54, 2072–2080.

Taylor, A., 1976. The cardiac responses to shell opening and closure in the bivalve *Arctica islandica* (L.). J. Exp. Biol. 64, 751–759.

Tran, D., Ciret, P., Ciutat, A., Durrieu, G., Massabuau, J.-C., 2003. Estimation of potential and limits of bivalve closure response to detect contaminants: application to cadmium. Environ. Toxicol. Chem. 22, 914–920.

Tran, D., Haberkorn, H., Soudant, P., Ciret, P., Massabuau, J.-C., 2010. Behavioral responses of *Crassostrea gigas* exposed to the harmful algae *Alexandrium minutum*. Aquaculture 98, 338–345.

Turner, R.L., 1970. Quicklime: effects on soft-bodied marine organisms. Science 168, 606–607.

Wood, F.B., 1908. Enemies and Perils of the Oyster Report State of Connecticut Shellfish Commissioners 1907-1908. Dec No 30, Appendix to Rept pp. 94–98.